

Case Studies

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[Practical]

Task 1: Create and Manage Files

Objective: Practice creating, listing, viewing, and deleting files

```
harshika@Harshika:~$ mkdir linux_task1
harshika@Harshika:~$ cd linux_task1
harshika@Harshika:~/linux_task1$ touch file1.txt file2.txt file3.txt file4.txt file5.txt
harshika@Harshika:~/linux_task1$ ls
file1.txt file2.txt file3.txt file4.txt file5.txt
harshika@Harshika:~/linux_task1$ ls -l
total 0
-rw-r--r-- 1 harshika harshika 0 Aug  6 16:40 file1.txt
-rw-r--r-- 1 harshika harshika 0 Aug  6 16:40 file2.txt
-rw-r--r-- 1 harshika harshika 0 Aug  6 16:40 file3.txt
-rw-r--r-- 1 harshika harshika 0 Aug  6 16:40 file4.txt
-rw-r--r-- 1 harshika harshika 0 Aug  6 16:40 file5.txt
harshika@Harshika:~/linux_task1$ nano file1.txt
harshika@Harshika:~/linux_task1$ cat file1.txt
This is my first Linux file.
harshika@Harshika:~/linux_task1$ rm file5.txt
harshika@Harshika:~/linux_task1$ ls
file1.txt file2.txt file3.txt file4.txt
harshika@Harshika:~/linux_task1$ |
```

Task 2: Copy and Move Files

Objective: Practice copying and moving files and directories

```

harshika@Harshika:~$ mkdir linux_task2
harshika@Harshika:~$ cd linux_task2
harshika@Harshika:~/linux_task2$ touch source.txt data.txt notes.txt
harshika@Harshika:~/linux_task2$ nano source.txt
harshika@Harshika:~/linux_task2$ mkdir backup
harshika@Harshika:~/linux_task2$ cp source.txt backup
harshika@Harshika:~/linux_task2$ ls backup
source.txt
harshika@Harshika:~/linux_task2$ cp data.txt backup/data_backup.txt
harshika@Harshika:~/linux_task2$ ls backup
data_backup.txt  source.txt
harshika@Harshika:~/linux_task2$ mv notes.txt backup/
harshika@Harshika:~/linux_task2$ ls
backup  data.txt  source.txt
harshika@Harshika:~/linux_task2$ ls backup
data_backup.txt  notes.txt  source.txt
harshika@Harshika:~/linux_task2$ mv source.txt original.txt
harshika@Harshika:~/linux_task2$ ls
backup  data.txt  original.txt
harshika@Harshika:~/linux_task2$ ls backup
data_backup.txt  notes.txt  source.txt
harshika@Harshika:~/linux_task2$ tree
.
├── backup
│   ├── data_backup.txt
│   ├── notes.txt
│   └── source.txt
├── data.txt
└── original.txt

2 directories, 5 files
harshika@Harshika:~/linux_task2$ |

```

Task 3: Working with head and tail

Objective: Practice viewing specific portions of a file.

```
harshika@Harshika:~$ mkdir linux_task3
harshika@Harshika:~$ cd linux_task3
harshika@Harshika:~/linux_task3$ touch students.txt
harshika@Harshika:~/linux_task3$ nano students.txt
harshika@Harshika:~/linux_task3$ cat students.txt
Ravinder
Srilatha
Varshika
Harshika
Santhvin
Chinni
Manu
Bhanu Priya
Nikshitha
Pranitha
Angela
Keerthi Priya
Jhansi Priya
Lahari
Sreedhar
harshika@Harshika:~/linux_task3$ head -5 students.txt
Ravinder
Srilatha
Varshika
Harshika
Santhvin
harshika@Harshika:~/linux_task3$ head students.txt
Ravinder
Srilatha
Varshika
Harshika
Santhvin
Chinni
Manu
Bhanu Priya
Nikshitha
Pranitha
harshika@Harshika:~/linux_task3$ tail -5 students.txt
Angela
Keerthi Priya
Jhansi Priya
```

```
Pranitha
Angela
Keerthi Priya
Jhansi Priya
Lahari
Sreedhar
harshika@Harshika:~/linux_task3$ head -5 students.txt
Ravinder
Srilatha
Varshika
Harshika
Santhvin
harshika@Harshika:~/linux_task3$ head students.txt
Ravinder
Srilatha
Varshika
Harshika
Santhvin
Chinni
Manu
Bhanu Priya
Nikshitha
Pranitha
harshika@Harshika:~/linux_task3$ tail -5 students.txt
Angela
Keerthi Priya
Jhansi Priya
Lahari
Sreedhar
harshika@Harshika:~/linux_task3$ tail students.txt
Chinni
Manu
Bhanu Priya
Nikshitha
Pranitha
Angela
Keerthi Priya
Jhansi Priya
Lahari
Sreedhar
harshika@Harshika:~/linux_task3$ |
```

Task 4: Organize Files

Objective: Organize files into different directories using mv

```
harshika@Harshika:~$ mkdir organization
harshika@Harshika:~$ cd organization
harshika@Harshika:~/organization$ touch file1.txt file2.txt file3.txt image1.jpg image2.jpg
harshika@Harshika:~/organization$ ls
file1.txt file2.txt file3.txt image1.jpg image2.jpg
harshika@Harshika:~/organization$ mkdir documents
harshika@Harshika:~/organization$ mkdir images
harshika@Harshika:~/organization$ mv *.txt documents/
harshika@Harshika:~/organization$ ls documents
file1.txt file2.txt file3.txt
harshika@Harshika:~/organization$ mv *.jpg images/
harshika@Harshika:~/organization$ ls images
image1.jpg image2.jpg
harshika@Harshika:~/organization$ ls
documents images
harshika@Harshika:~/organization$ mv documents/file1.txt documents/notes.txt
harshika@Harshika:~/organization$ ls documents
file2.txt file3.txt notes.txt
harshika@Harshika:~/organization$ mv images/image1.jpg images/photo.jpg
harshika@Harshika:~/organization$ ls images
image2.jpg photo.jpg
harshika@Harshika:~/organization$ tree
.
├── documents
│   ├── file2.txt
│   ├── file3.txt
│   └── notes.txt
└── images
    ├── image2.jpg
    └── photo.jpg

3 directories, 5 files
harshika@Harshika:~/organization$ |
```

Task 5: Directory Management

Objective: Practice creating nested directories and removing them

```

harshika@Harshika:~$ mkdir Project
harshika@Harshika:~$ cd Project
harshika@Harshika:~/Project$ mkdir source/
harshika@Harshika:~/Project$ mkdir source/c/
harshika@Harshika:~/Project$ mkdir source/python/
harshika@Harshika:~/Project$ mkdir docs/
harshika@Harshika:~/Project$ mkdir backup/
harshika@Harshika:~/Project$ touch source/c/file1.c
harshika@Harshika:~/Project$ touch source/c/file2.c
harshika@Harshika:~/Project$ touch source/c/file3.c
harshika@Harshika:~/Project$ ls source/c/
file1.c file2.c file3.c
harshika@Harshika:~/Project$ touch source/python/program1.py source/python/program2.py
harshika@Harshika:~/Project$ ls source/python/
program1.py program2.py
harshika@Harshika:~/Project$ tree
.
├── backup
├── docs
├── source
│   ├── c
│   │   ├── file1.c
│   │   ├── file2.c
│   │   └── file3.c
│   └── python
│       ├── program1.py
│       └── program2.py

```

```

6 directories, 5 files
harshika@Harshika:~/Project$ rmdir backup
harshika@Harshika:~/Project$ ls
docs source
harshika@Harshika:~/Project$ mkdir backup
harshika@Harshika:~/Project$ touch backup/data.txt
harshika@Harshika:~/Project$ rmdir backup
rmdir: failed to remove 'backup': Directory not empty
harshika@Harshika:~/Project$ rm backup/data.txt
harshika@Harshika:~/Project$ ls backup
harshika@Harshika:~/Project$ rmdir backup
harshika@Harshika:~/Project$ ls
docs source

```

```

harshika@Harshika:~/Project$ mkdir backup
harshika@Harshika:~/Project$ touch backup/data.txt
harshika@Harshika:~/Project$ rmdir backup
rmdir: failed to remove 'backup': Directory not empty
harshika@Harshika:~/Project$ rm backup/data.txt
harshika@Harshika:~/Project$ ls backup
harshika@Harshika:~/Project$ rmdir backup
harshika@Harshika:~/Project$ ls
docs source
harshika@Harshika:~/Project$ tree

```

```

.
├── docs
├── source
│   ├── c
│   │   ├── file1.c
│   │   ├── file2.c
│   │   └── file3.c
│   └── python
│       ├── program1.py
│       └── program2.py

```

```

5 directories, 5 files
harshika@Harshika:~/Project$ |

```